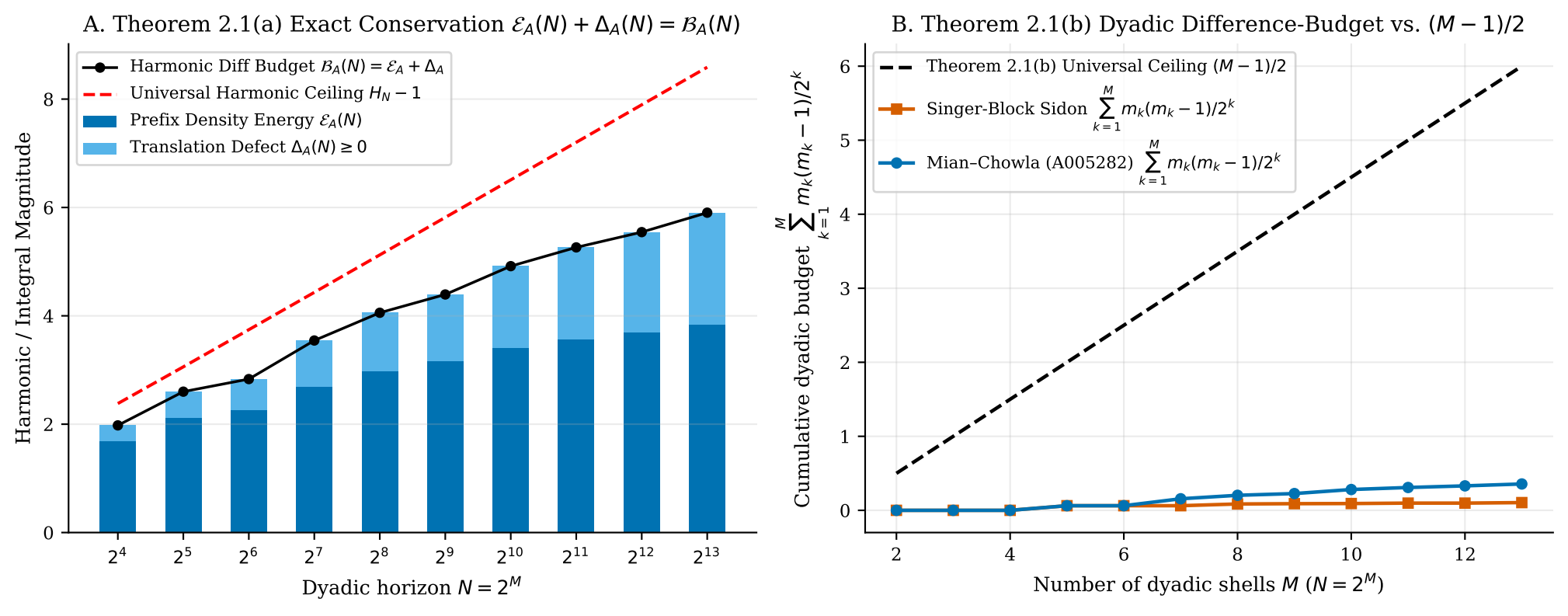
A. Theorem 2.1(a) Exact Conservation $\mathcal{E}_A(N) + \Delta_A(N) = \mathcal{B}_A(N)$
Harmonic Diff Budget $\mathcal{B}_A(N) = \mathcal{E}_A + \Delta_A$
Universal Harmonic Ceiling $H_N - 1$
Prefix Density Energy $\mathcal{E}_A(N)$
Translation Defect $\Delta_A(N) \geq 0$
Harmonic / Integral Magnitude
0
2
4
6
8
2^4
2^5
2^6
2^7
2^8
2^9
2^{10}
2^{11}
2^{12}
2^{13}
Dyadic horizon $N = 2^M$
B. Theorem 2.1(b) Dyadic Difference-Budget vs. $(M-1)/2$
Theorem 2.1(b) Universal Ceiling $(M-1)/2$
Singer-Block Sidon $\sum_{k=1}^{M} m_k(m_k-1)/2^k$
Mian–Chowla (A005282) $\sum_{k=1}^{M} m_k(m_k-1)/2^k$
Cumulative dyadic budget $\sum_{k=1}^{M} m_k(m_k-1)/2^k$
0
1
2
3
4
5
6
2
4
6
8
10
12
Number of dyadic shells M $(N = 2^M)$